Programming for Problem Solving Using Python
Unit-1

Session 2: Introduction to Anaconda, Jupyter Installation Guide

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Session-1

Session-1: Introduction to Anaconda, Jupyter Installation Guide

Session-2: Anaconda Python

- Data science libraries- Pandas, NumPy and Dask.
- Visualization libraries- Datashader, matplotlib and Holoviews.
- Jupyter Notebook- a shareable notebook combining live code, visualizations and text.

Session-2: Anaconda Python

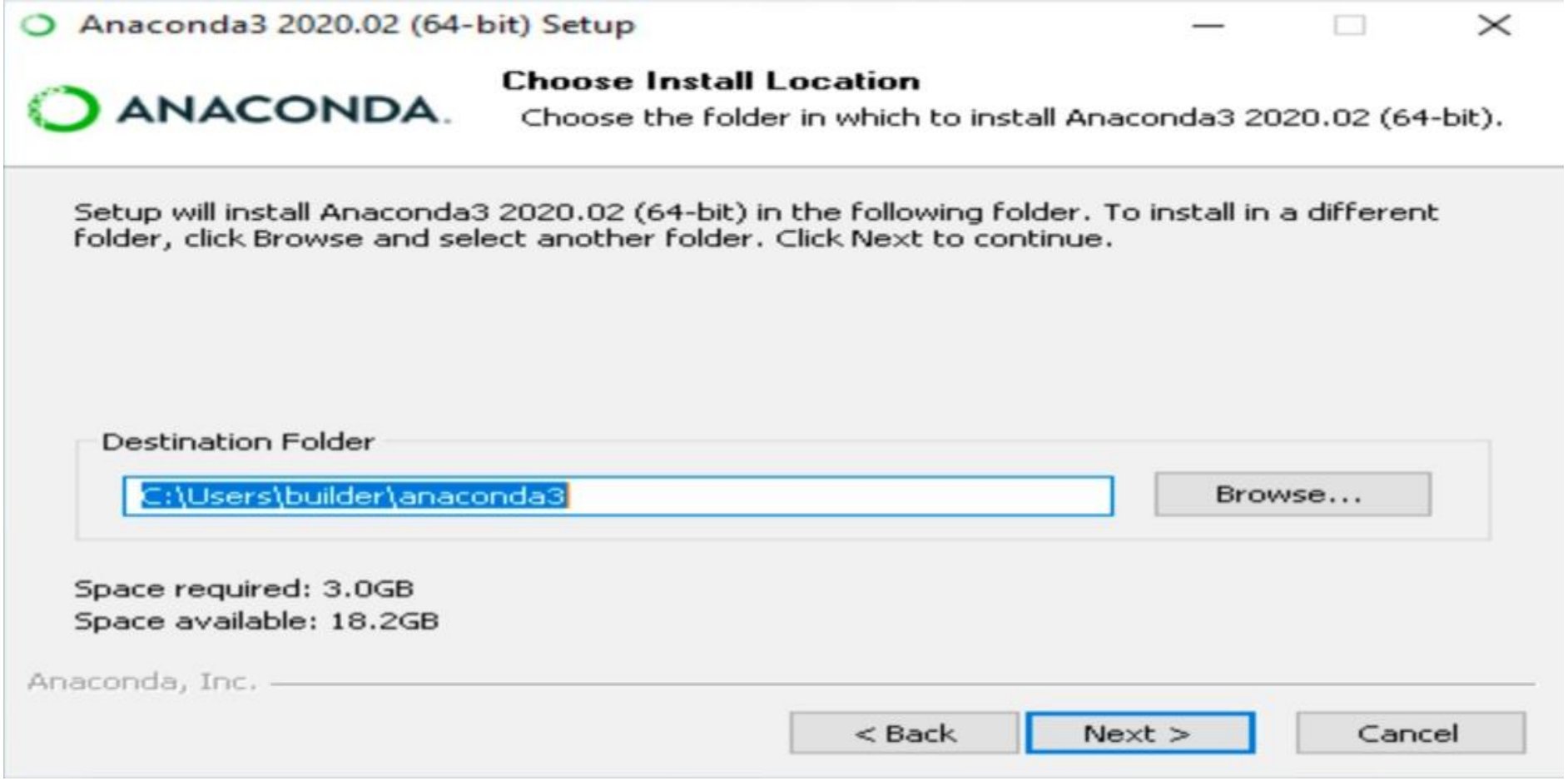
- Anaconda- open-source software containing Jupyter, spyder, etc used for data analytics.
- Works for R and Python.
- Package versions are managed by the package management system called Conda.
- Machine learning libraries- Tensor Flow, sci-kit-learn and Theano.

Session-2: Popular Python IDE's

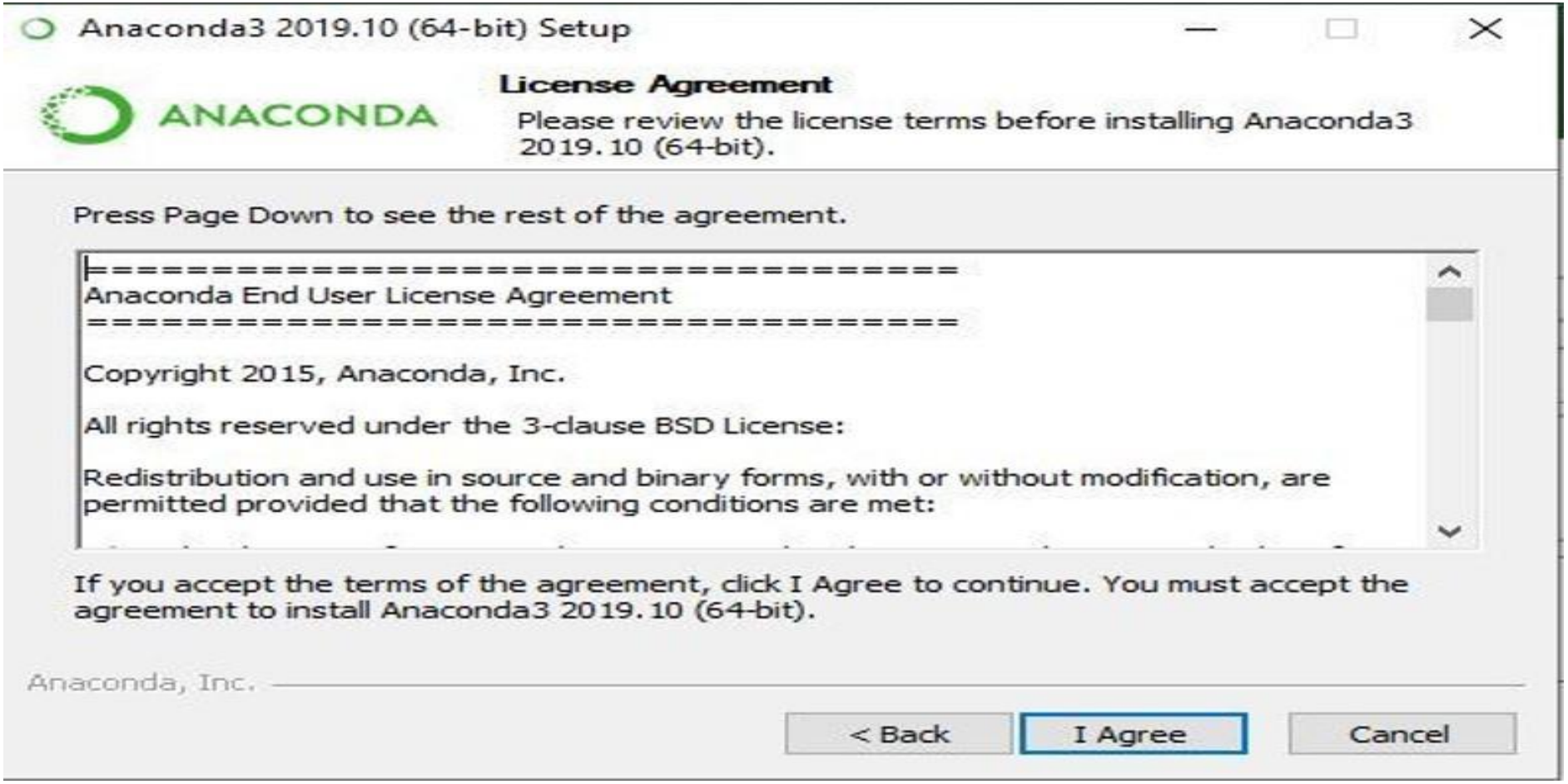
Session-2: Steps to Install Anaconda Python

- Step 1: Download Anaconda
- Go to Anaconda.com
- https://www.anaconda.com/products/individual
- Anaconda Installers
  1. https://repo.anaconda.com/archive/Anaconda3-2020.07-Windows-x86_64.exe
  2. https://repo.anaconda.com/archive/Anaconda3-2020.07-Windows-x86.exe

Session-2: Steps to Install Anaconda Python



Session-2: Steps to Install Anaconda Python



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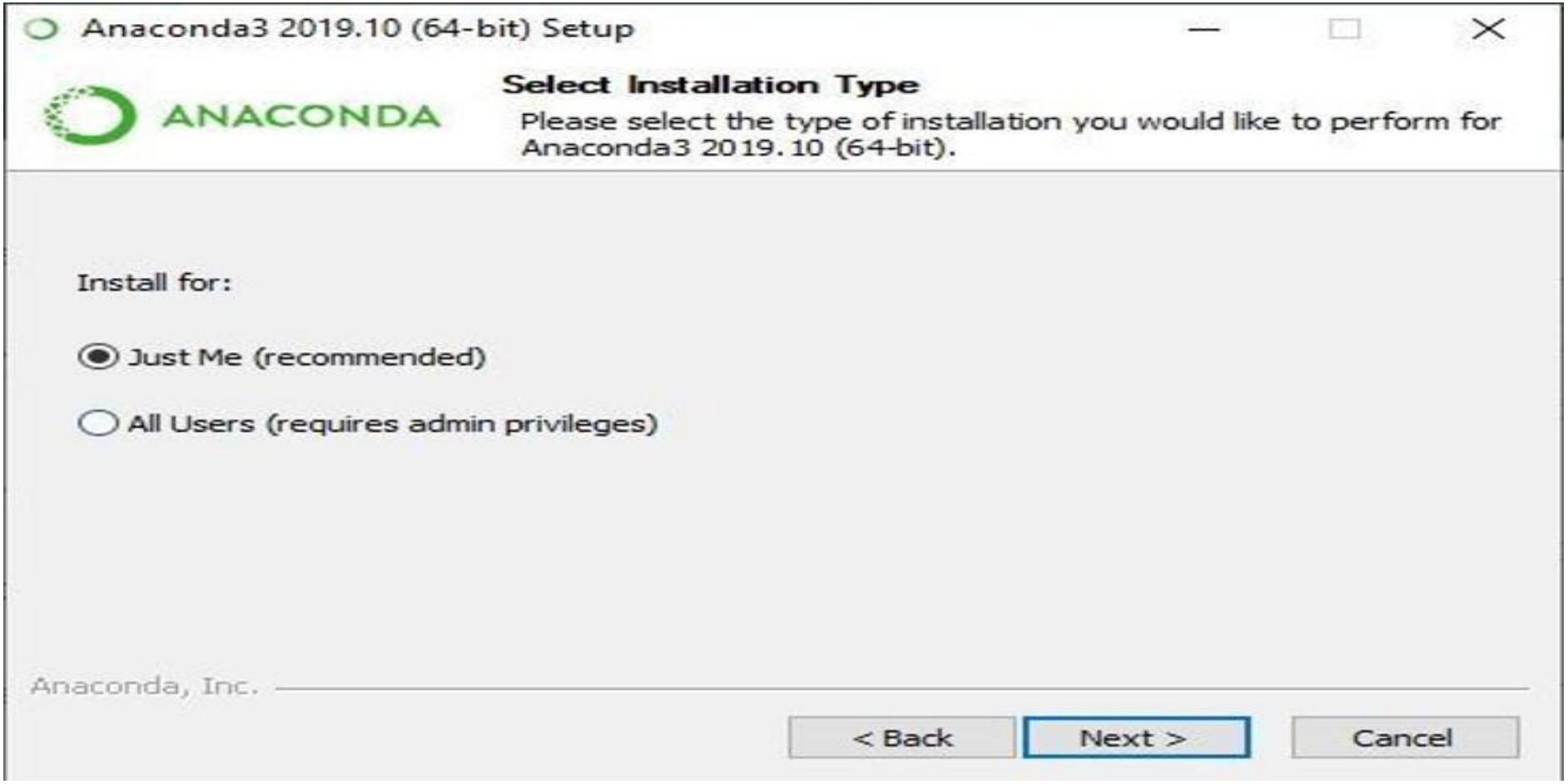
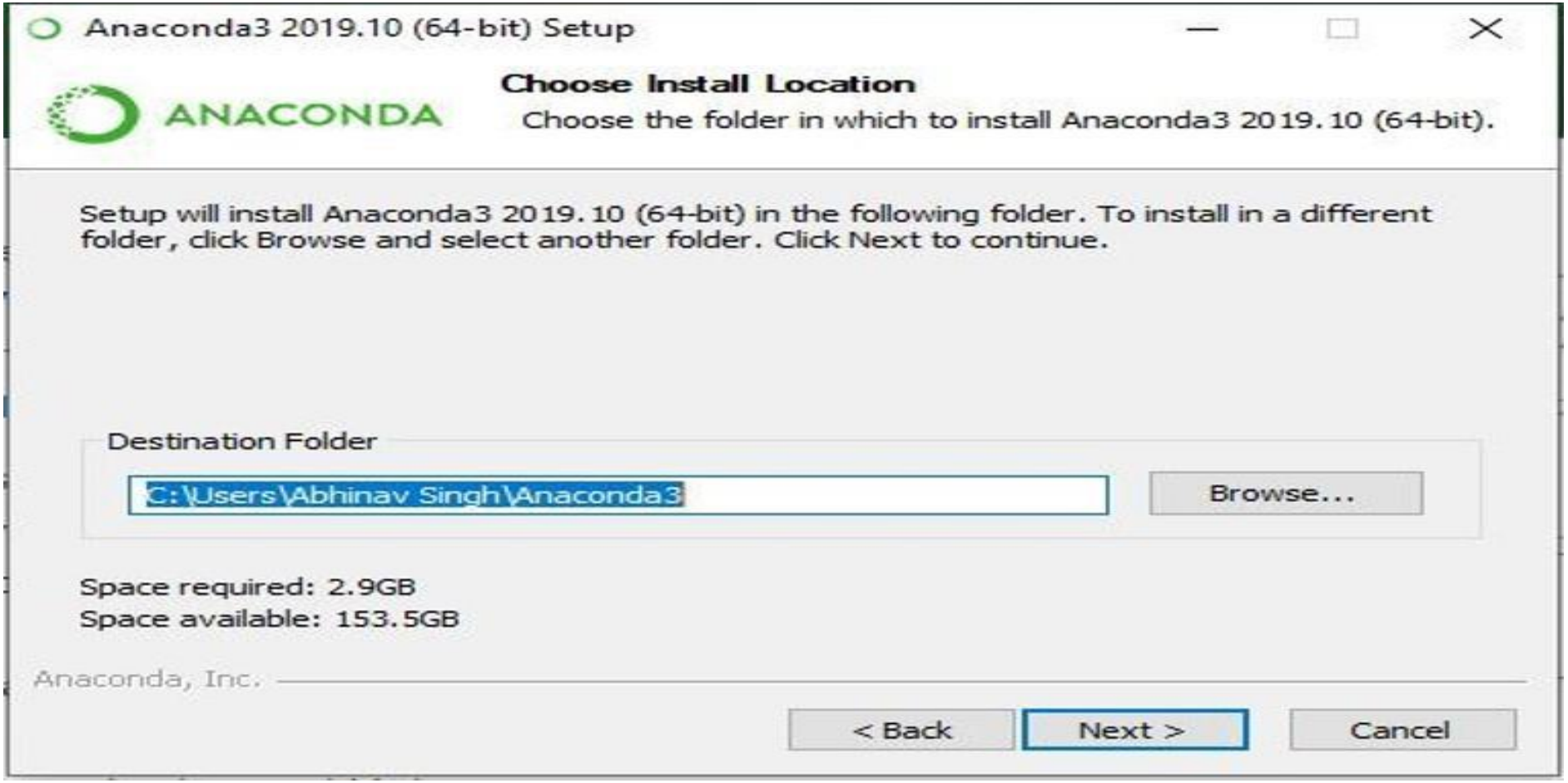


Session-2: Steps to Install Anaconda Python



Session-2: Steps to Install Anaconda Python

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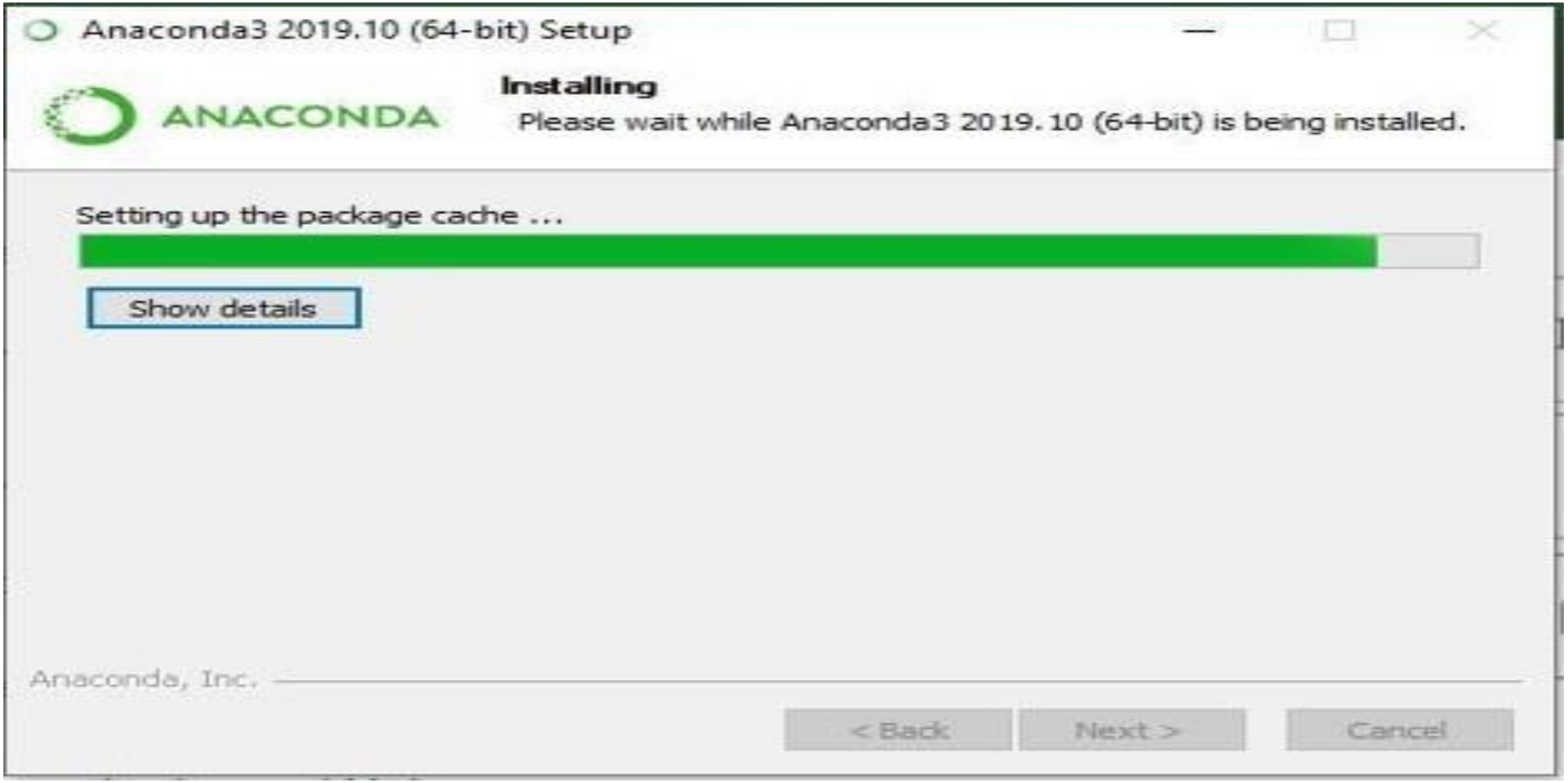
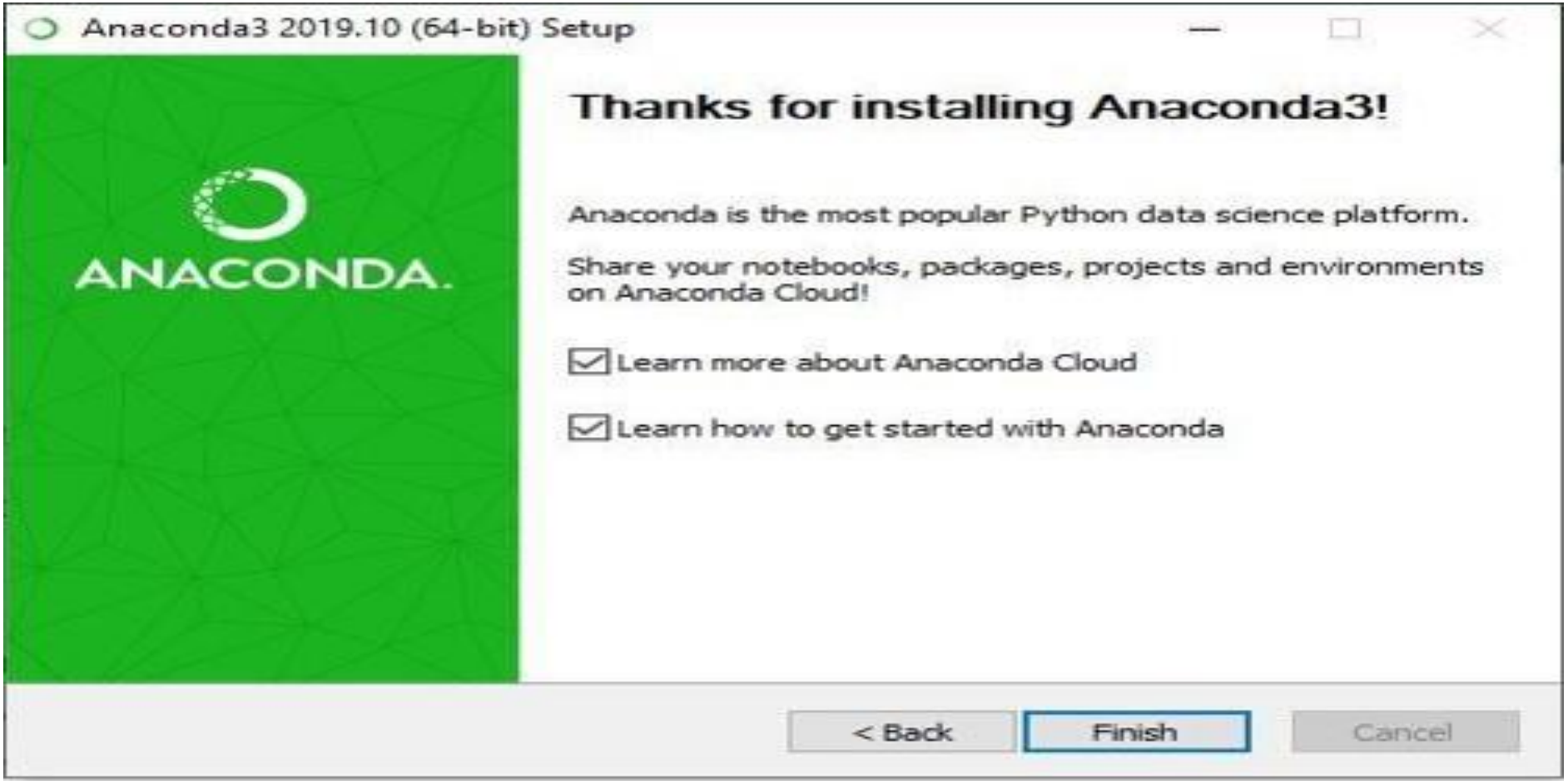
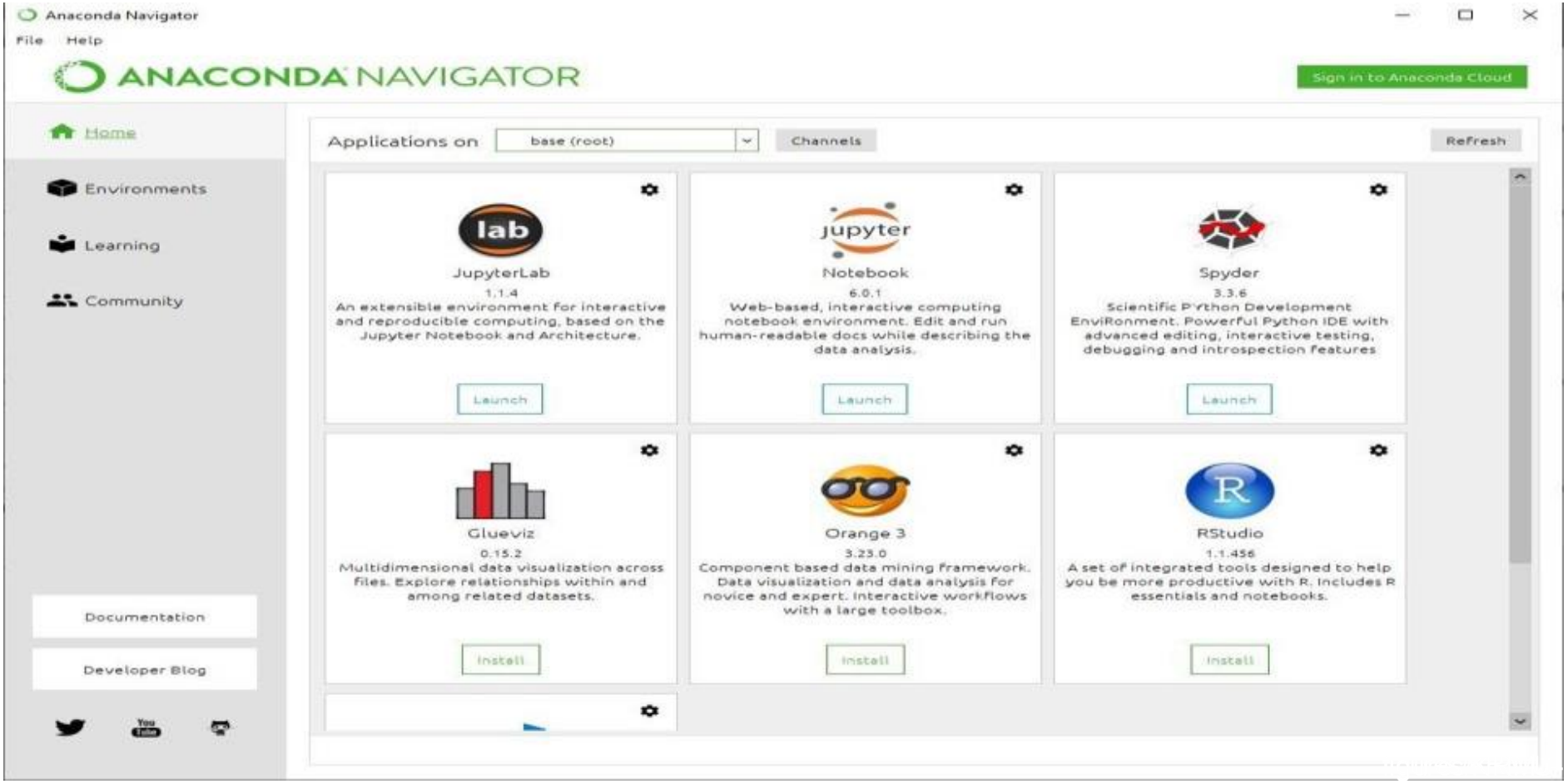


Session-2: Steps to Install Anaconda Python



Session-2: Steps to Install Anaconda Python



Session-2: Jupyter

The Jupyter Notebook is an open source web application that you can use to create and share documents that contain live code, equations, visualizations, and text.

Jupyter supports the core supported programming languages like: Julia, **Python**, and R.

Session-2: Jupyter Installation

- Install Python and Jupyter using the Anaconda Distribution, which includes Python, the Jupyter Notebook, and other commonly used packages for scien- tific computing and data science.
- You can download Anaconda's latest Python3 version.
- Installing Jupyter Notebook using pip:
  - python3 -m pip install –upgrade pip
  - python3 -m pip install jupyter

Session-2: Working with Jupyter Notebook

- To start the jupyter notebook, type the below com- mand in the terminal.
  jupyter notebook

- This will print some information about the notebook server in your terminal, including the URL of the web application (by default, http://localhost:8888) and then open your default web browser to this URL.

Session-2: Working with Jupyter Notebook

nikhil@nikhil-Lenovo-V130-15IKB: ~/Desktop/gfg

File Edit View Search Terminal Help

```
nikhil@nikhil-Lenovo-V130-15IKB:~/Desktop/gfg$ jupyter notebook
[I 18:27:32.989 NotebookApp] Serving notebooks from local directory: /home/nikhil/Desktop/gfg
[I 18:27:32.989 NotebookApp] The Jupyter Notebook is running at:
[I 18:27:32.989 NotebookApp] http://localhost:8888/?token=920909f0ba958cee4e6a9ae5644b43c73f5
44cc07cbe2846
[I 18:27:32.989 NotebookApp]  or http://127.0.0.1:8888/?token=920909f0ba958cee4e6a9ae5644b43c
73f544cc07cbe2846
[I 18:27:32.989 NotebookApp] Use Control-C to stop this server and shut down all kernels (twi
ce to skip confirmation).
[C 18:27:32.997 NotebookApp]

    To access the notebook, open this file in a browser:
        file:///home/nikhil/.local/share/jupyter/runtime/nbserver-9407-open.html
    Or copy and paste one of these URLs:
        http://localhost:8888/?token=920909f0ba958cee4e6a9ae5644b43c73f544cc07cbe2846
     or http://127.0.0.1:8888/?token=920909f0ba958cee4e6a9ae5644b43c73f544cc07cbe2846
Opening in existing browser session.
```

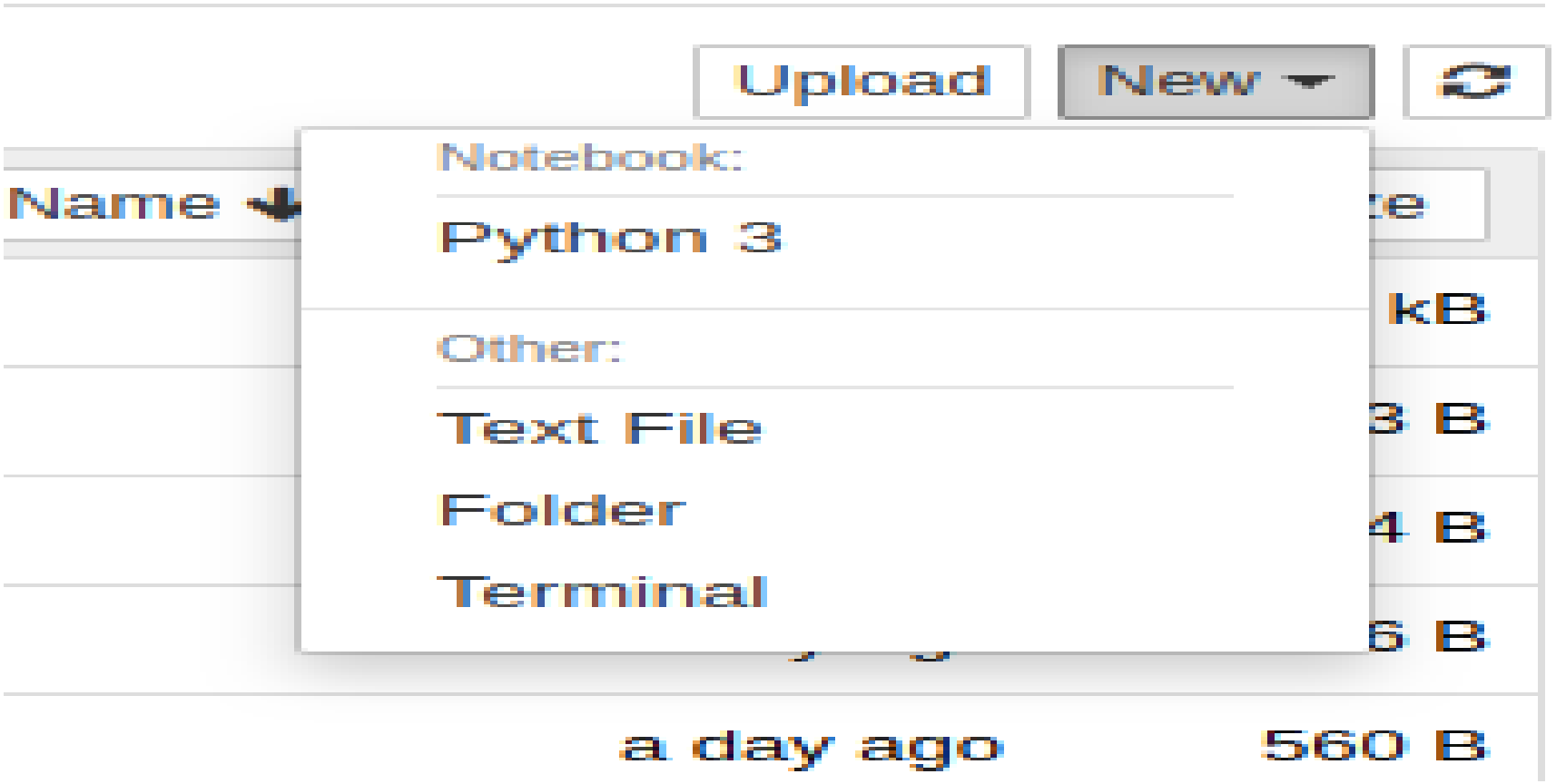
figures/KRIWU-

Session-2: Creating a Jupyter Notebook

Creating a Notebook

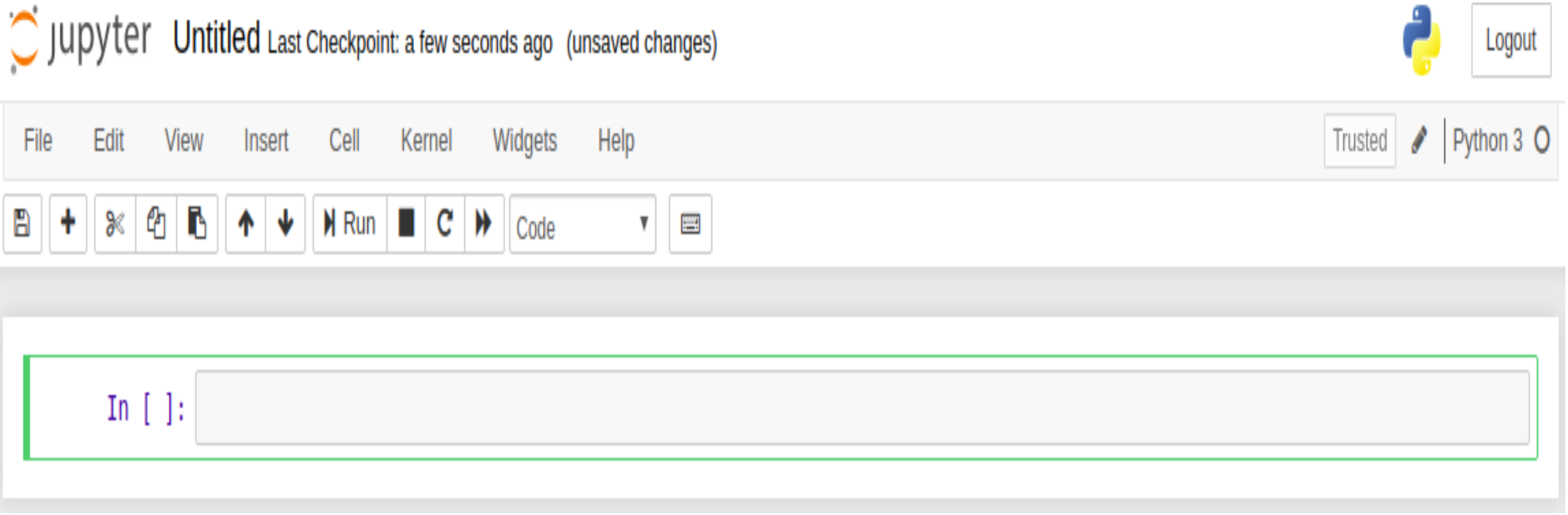
- To create a new notebook, click on the new button at the top right corner. Click it to open a drop-down list and then if you'll click on Python3, it will open a new notebook.

Session-2: Creating a Jupyter Notebook



Session-2: Creating a Jupyter Notebook

The web page should look like this:



Session-2: Spyder

Spyder is an open source cross-platform integrated development environment (IDE) for scientific programming in the Python language.

Session-2: Spyder Installation

- **Prerequisites**
  Before we start, ensure that you have Python and pip installed on your system. Spyder requires Python 3.7 or later.
- **Step 1: Install Spyder**
  To install Spyder without Anaconda, we'll use pip, the Python package installer. Open your terminal or command prompt and type the following command:
  pip install spyder
  This command will download and install Spyder and its dependencies.

Session-2: Spyder Installation

- **Step 2: Install PyQt5**
  Spyder uses PyQt5 for its graphical interface. If it's not already installed, you can add it with pip:

  pip install pyqt5
- **Step 3: Install Spyder Kernels**
  Spyder kernels are necessary for running and interacting with your code. Install them with the following command:

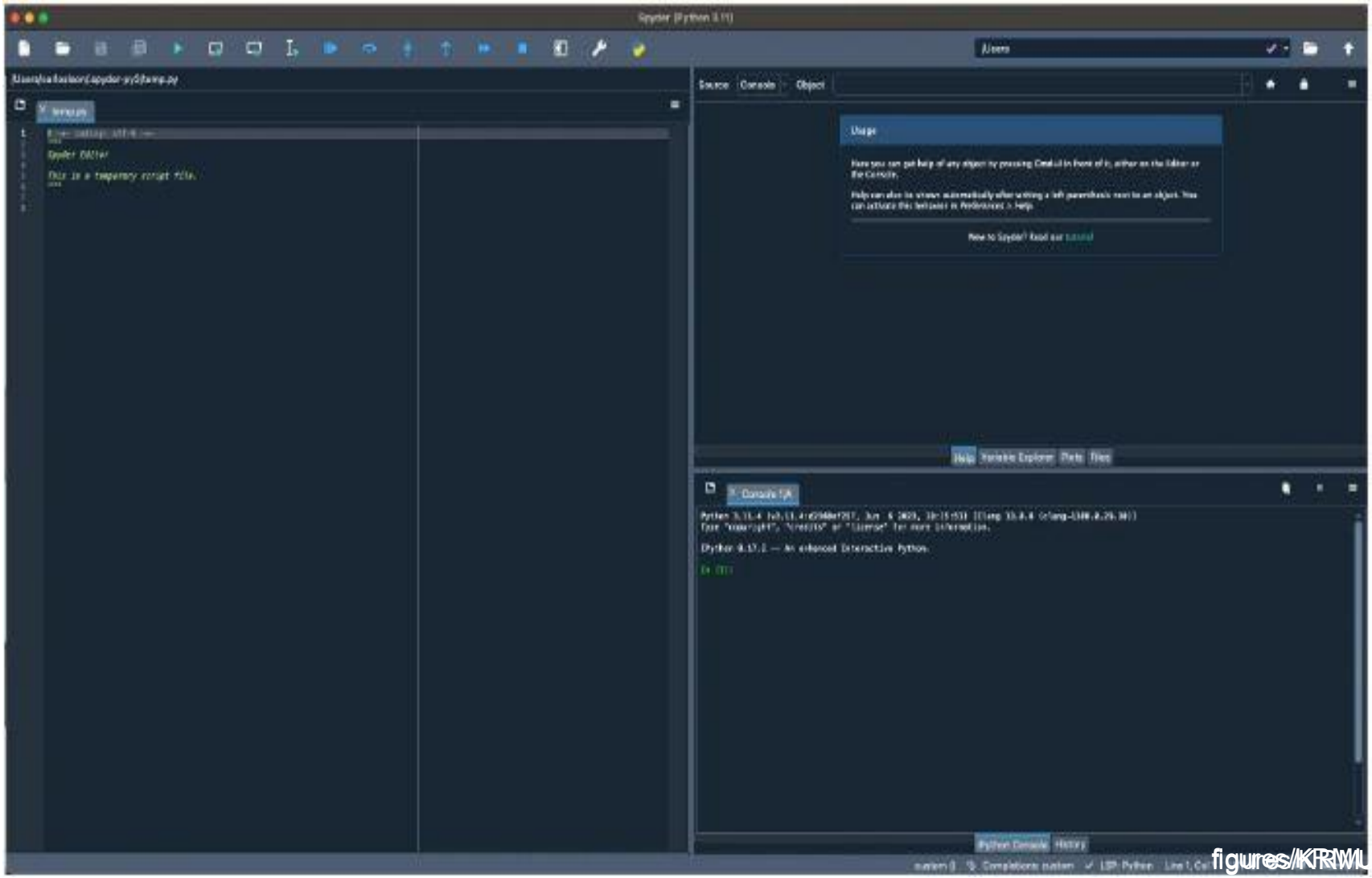
  pip install spyder-kernels

Session-2: Spyder Installation

- **Step 4: Launch Spyder**
  Now that everything is installed, you can launch Spyder by typing spyder in your terminal or command prompt:
  spyder

  If everything has been installed correctly, the Spyder IDE should open.

Session-1: Spyder Installation



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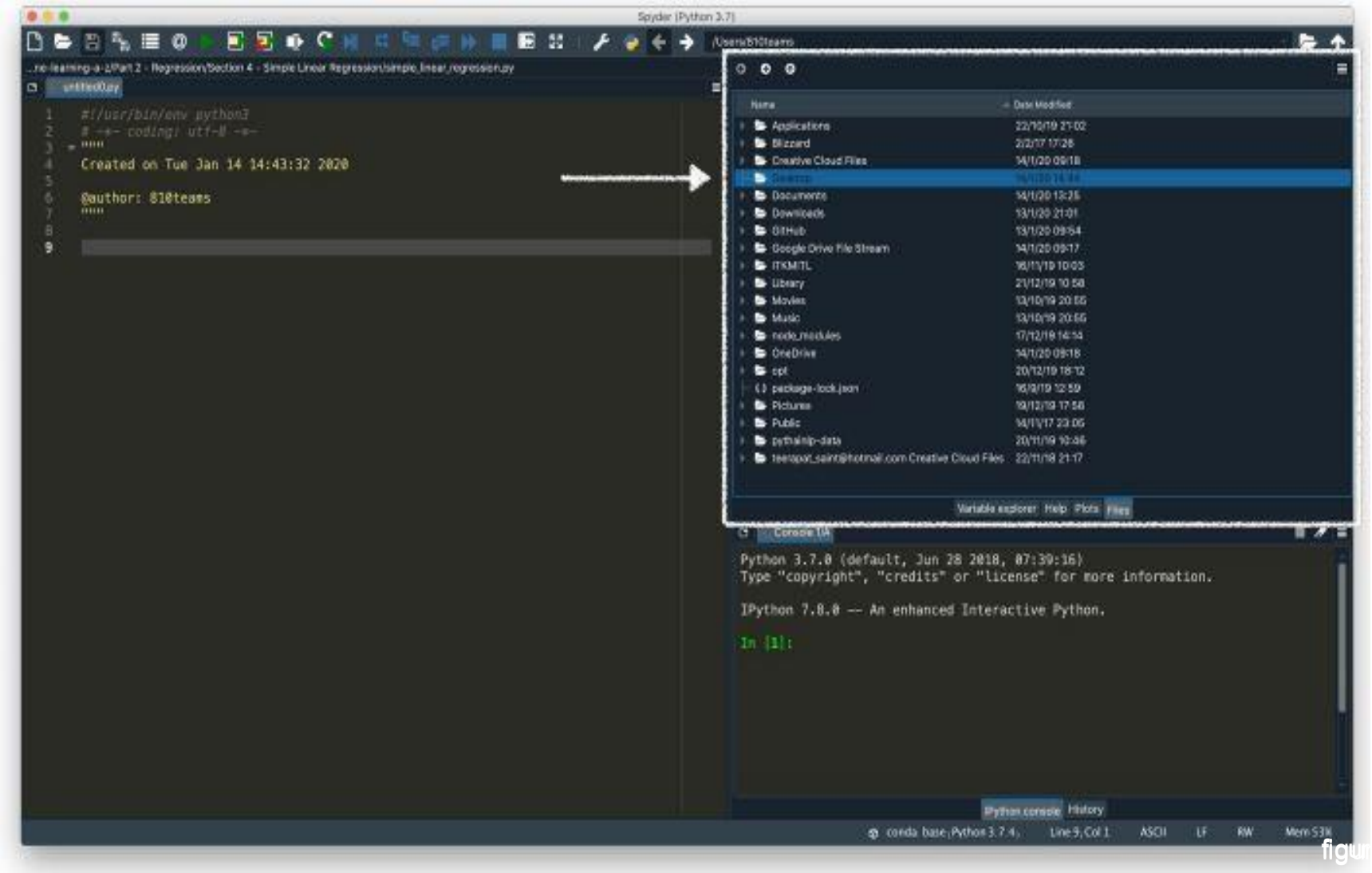
Session-2: Working with Spyder

File Explorer

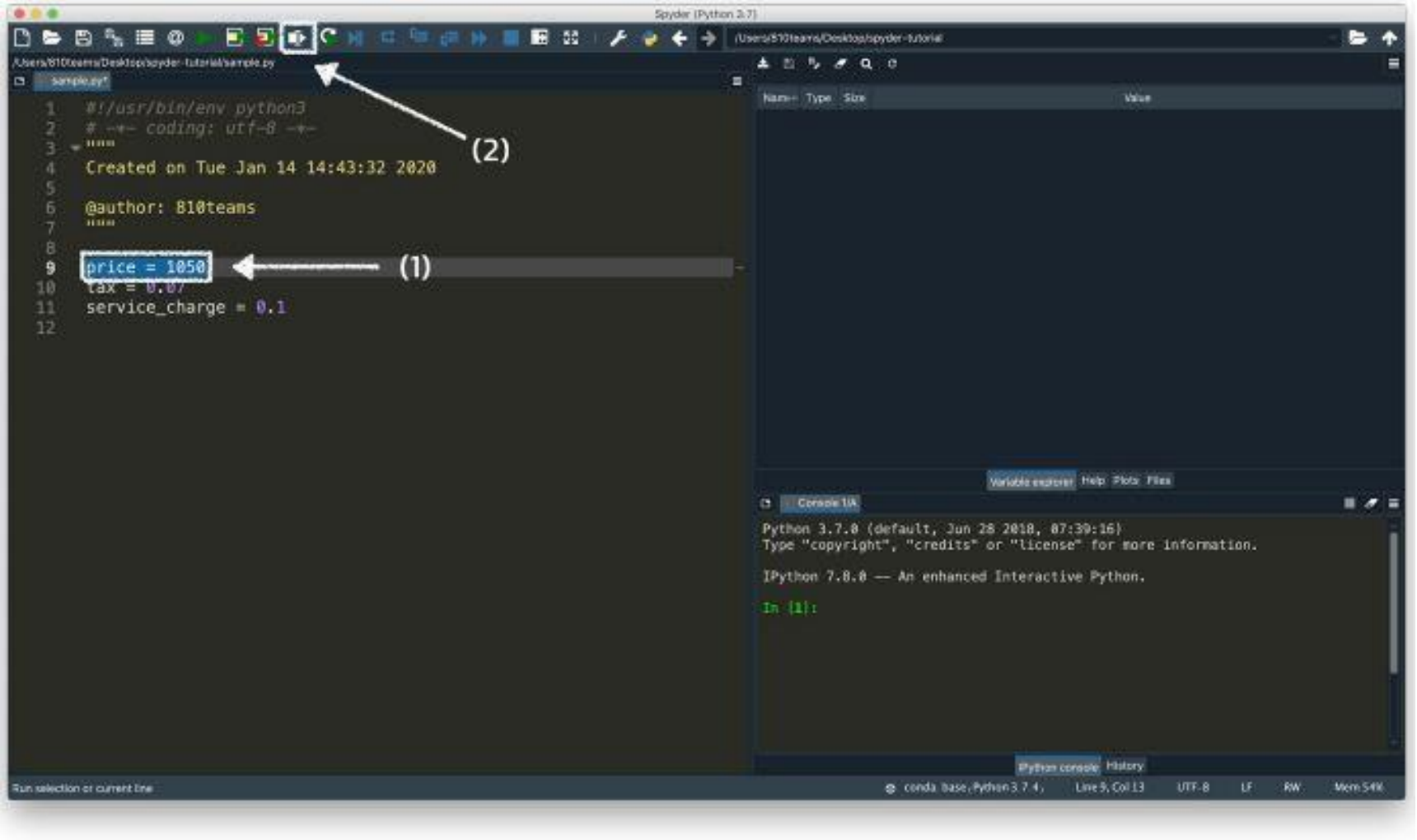
Explore:- Use file explorer to explore files and folders in your system.

Set as workspace:- Double click the folder to set the folder as the workspace. Running scripts will be done in the selected directory.

Session-2: Working with Spyder

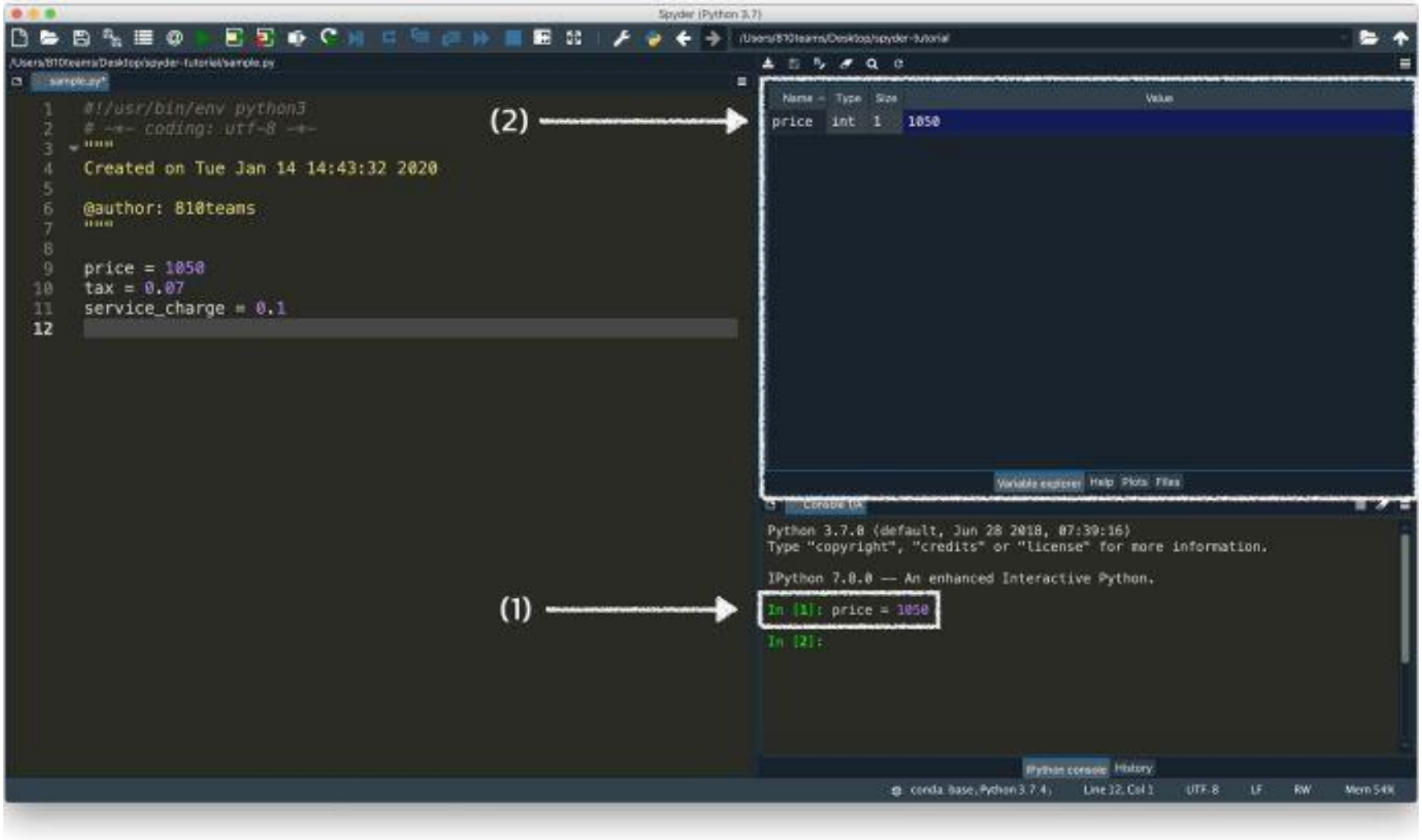


Session-2: Working with Spyder



Run the highlighted part of code

Session-1: Working with Spyder



Executed variable declaration appears in the variable explorer

Session-2: PIP Command in python

- Standard package manager for Python.
- Used for installing libraries and packages not available in the Python standard library.
- Comes pre-installed in Python version 3.4 and above.

Session-2: PIP Command in python

- To check the installed version of pip:-
  pip –version
  pip install some-package-name

- Use the list command to list all the packages installed on your system.
  pip list
  pip install pandas
  pip show pandas
  pip uninstall pandas

Session-2: Install PIP on Windows

Prerequisites for Pip Installation:-

- Check if pip is already installed by running the following on the command line:

  pip –version or pip -V

  If pip is installed, you'll see something similar to the following output:

  C:usersjdoe>pip-- versionPip19.2.3from c:usersjdoeappdatalocalprogramspython python38-32libsite-packagespip(python3.8)

Session-2: Install PIP on Windows

2. Verify that Python is installed. If pip is not installed, you can confirm that Python is available on your local machine and determine the version by running the following command:
   python –version

3. You should see something like the following if Python is installed:
   ```
   C:usersjdoe>pip--versionPython3.8.0
   ```

Session-2: Install PIP on Windows

Pip Installation:-

4. Download PIP get-pip.py
curl https://bootstrap.pypa.io/get-pip.py -o get-pip.py

```
C:\Users\PC>curl https://bootstrap.pypa.io/get-pip.py -o get-pip.py
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 2570k  100 2570k    0     0  7138k      0 --:--:-- --:--:-- --:--:-- 7180k
```

Session-2: Install PIP on Windows

5 Installing PIP on Windows
To install PIP, run the following Python command:

python get-pip.py

```
C:\Users\PC>python get-pip.py
Defaulting to user installation because normal site-packages is not writeable
Collecting pip
  Using cached pip-23.3.1-py3-none-any.whl.metadata (3.5 kB)
Using cached pip-23.3.1-py3-none-any.whl (2.1 MB)
Installing collected packages: pip
  WARNING: The scripts pip.exe, pip3.11.exe and pip3.exe are installed in 'C:\Users\PC\AppData\Local\Packages\PythonSoft
wareFoundation.Python.3.11_qbz5n2kfra8p0\LocalCache\local-packages\Python311\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed pip-23.3.1
```

Session-2: Install PIP on Windows

6 Verify Installation

python -m pip help

If PIP is installed, the program runs without any er- rors. The output shows the software's location and a list of pip commands.

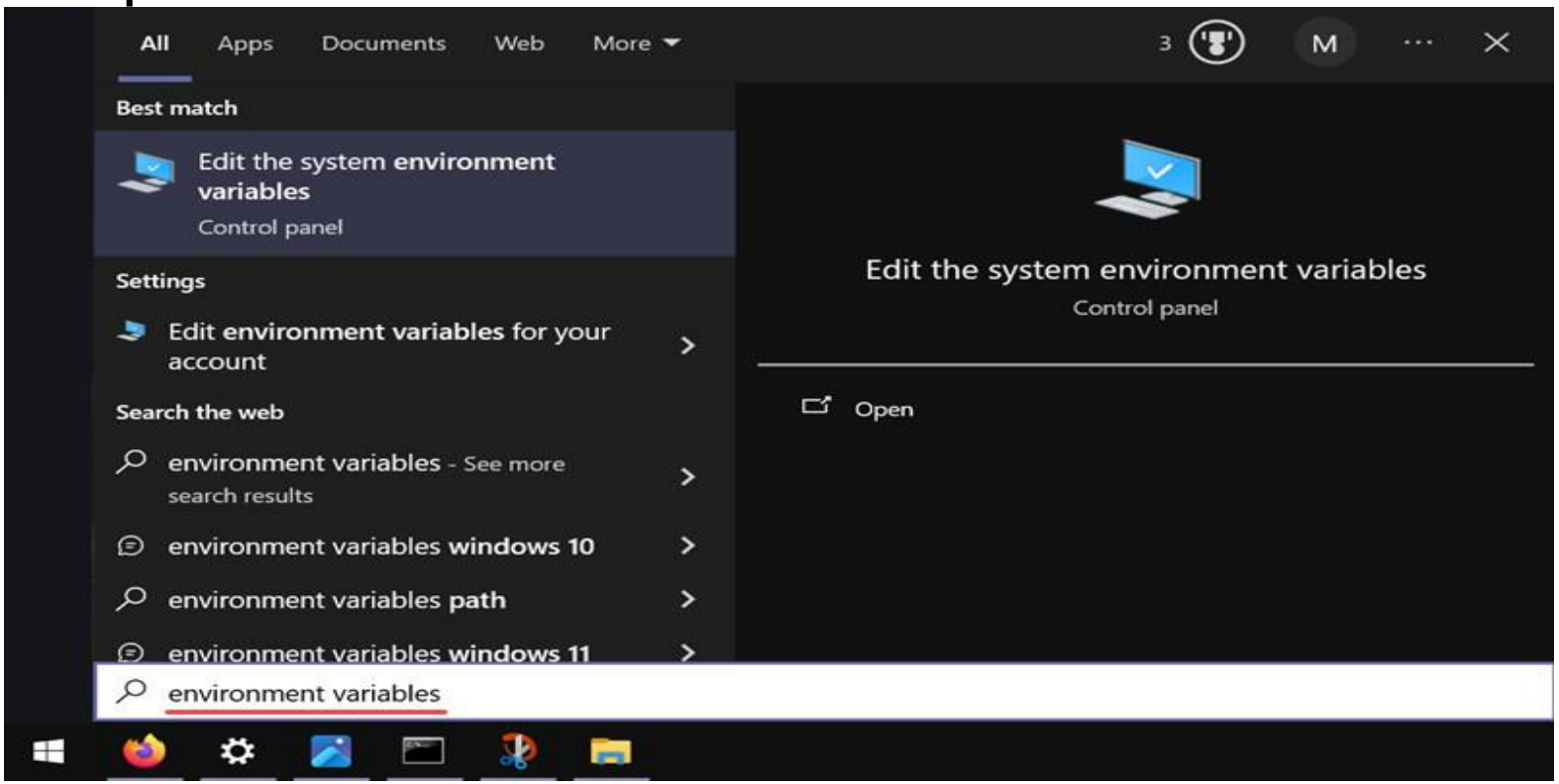
```
C:\Users\PC>python -m pip help

Usage:
  C:\Users\PC\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.11_

Commands:
  install                     Install packages.
  download                    Download packages.
  uninstall                   Uninstall packages.
  freeze                      Output installed packages in requirements format.
  inspect                     Inspect the python environment.
  list                        List installed packages.
  show                        Show information about installed packages.
  check                       Verify installed packages have compatible dependencies.
  config                      Manage local and global configuration.
  search                      Search PyPI for packages.
```

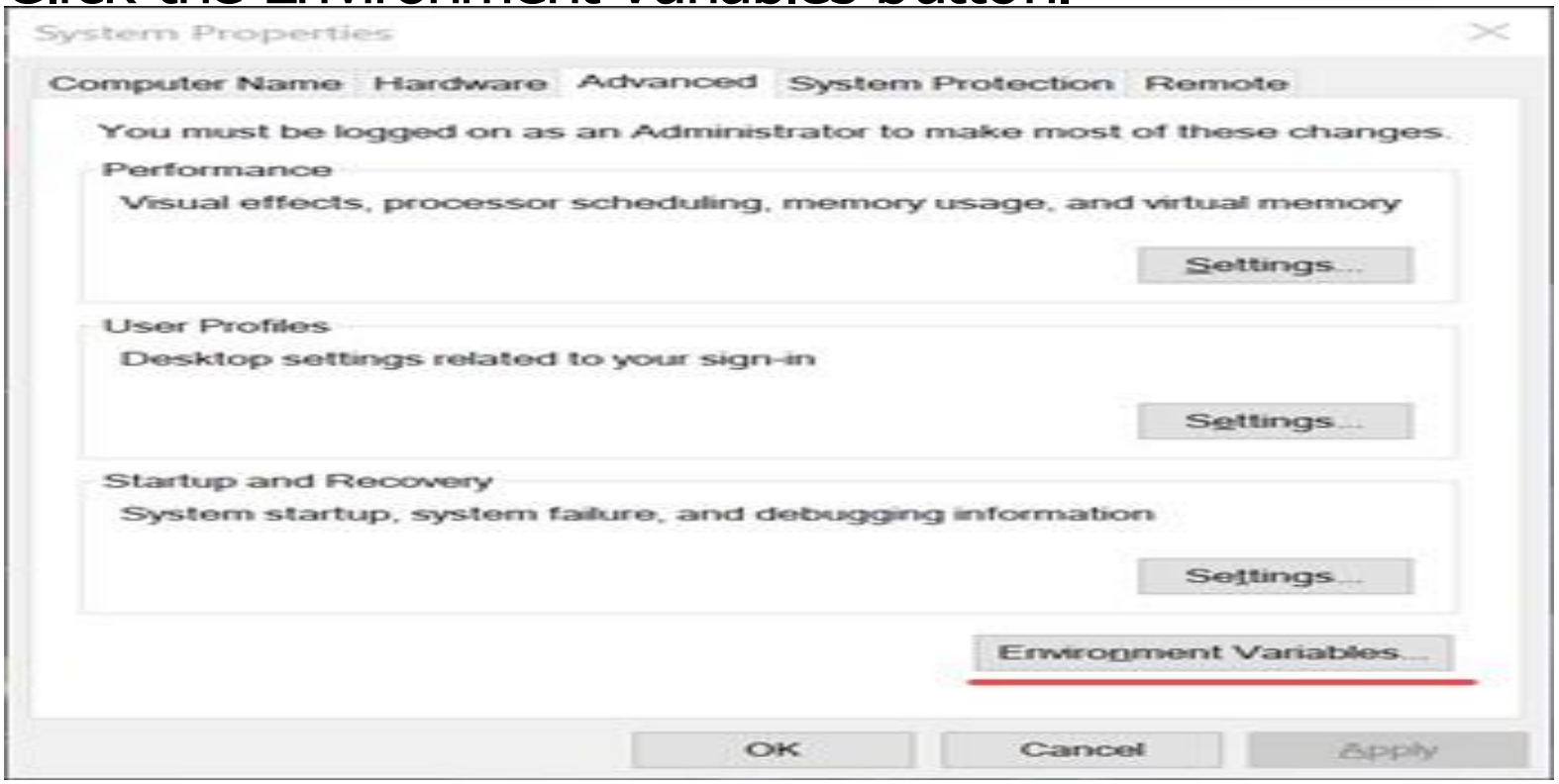
Session-2: Install PIP on Windows

7. Add Pip to Path
   - Open the Start menu, search for Environment Variables, and press Enter.



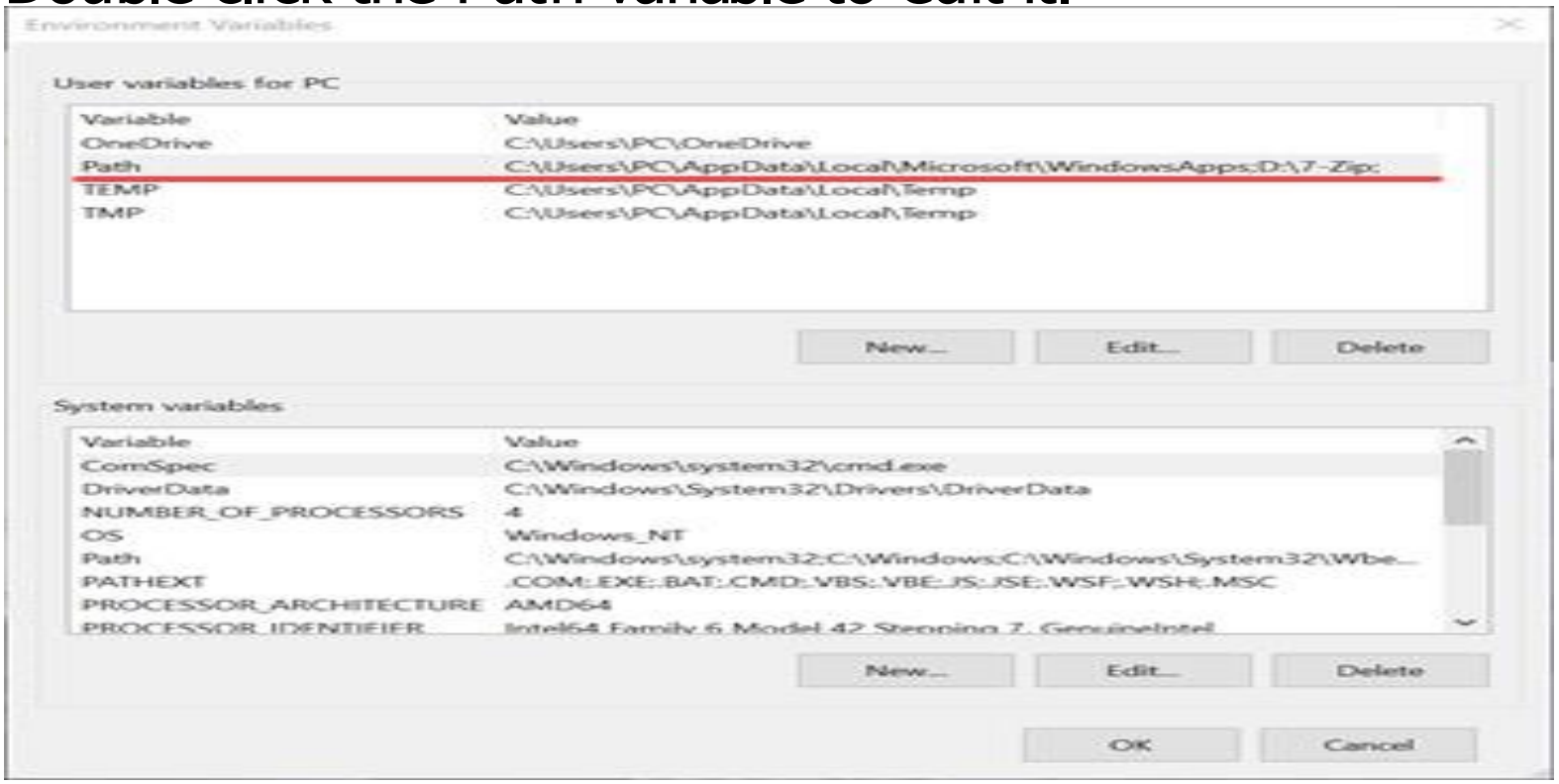
Session-2: Install PIP on Windows

- Click the Environment Variables button.



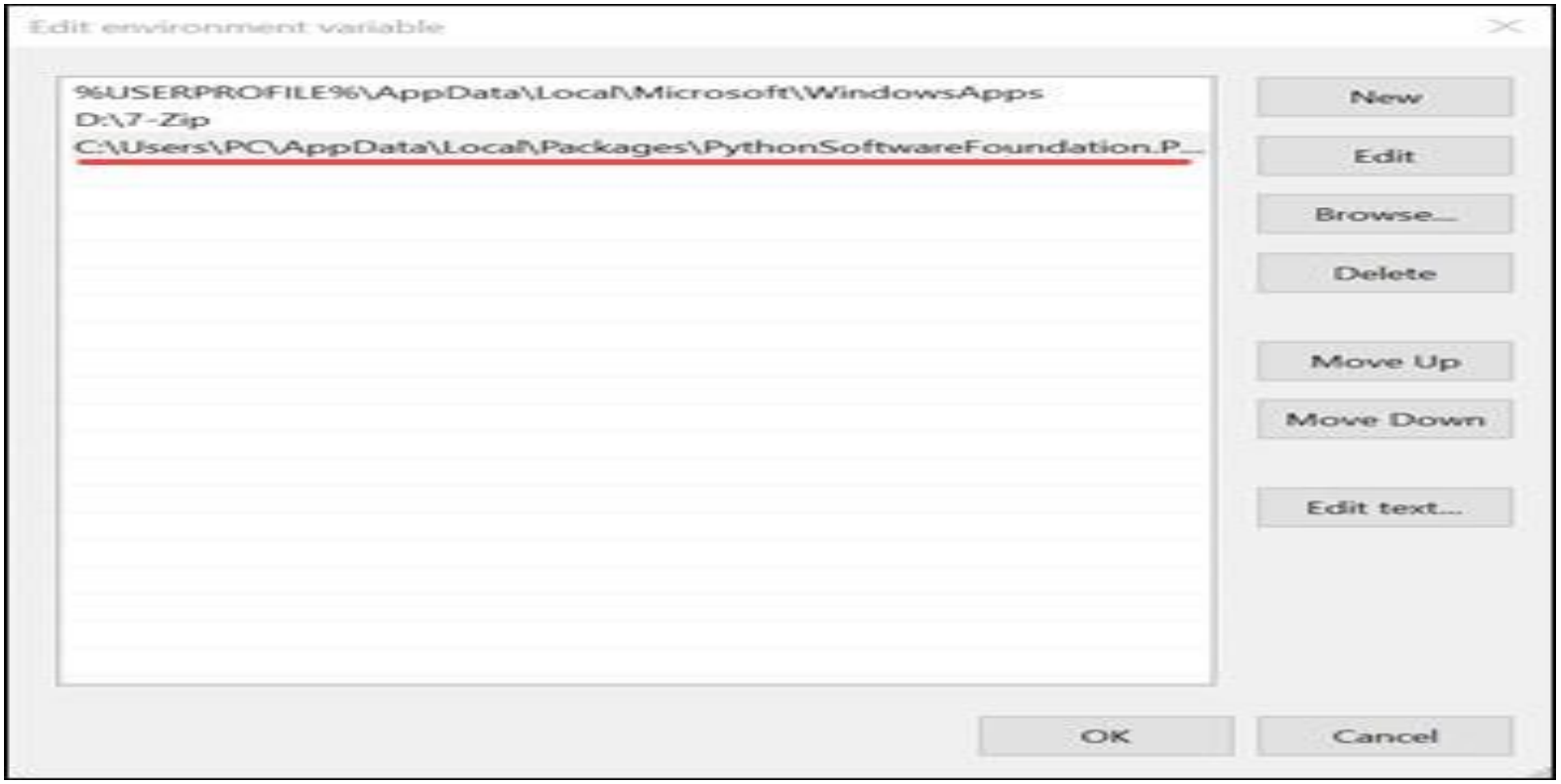
Session-2: Install PIP on Windows

- Double-click the Path variable to edit it.



Session-2: Install PIP on Windows

- Select New and add the directory where PIP is installed.



Session-1: Install PIP on Windows

- Click OK to save the changes.
- Open a new command prompt session and run the following command to test the changes:
  pip help

```
C:\Users\PC>pip help

Usage:
  C:\Users\PC\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.11_

Commands:
  install                     Install packages.
  download                    Download packages.
  uninstall                   Uninstall packages.
  freeze                      Output installed packages in requirements format.
  inspect                     Inspect the python environment.
  list                        List installed packages.
  show                        Show information about installed packages.
  check                       Verify installed packages have compatible dependencies.
```

Session-2: Python Libraries and Packages for Beginners

- **Matplotlib:** 2-dimensional graphs and plots.
- **Numpy:** is a popular array–processing package of Python.
- **OpenCV:** Open Source Computer Vision is a python package for image processing.
- **Requests:** is a rich Python HTTP library.

Session-2: Python Libraries and Packages for Beginners

- **Keras:** People who want to learn deep neural networks.
- **TensorFlow:** is a free, open-source python machine learning library.
- **NLTK:** Natural language toolkit is one of the most popular Python NLP libraries

Session-2: Python Libraries and Packages for Beginners

- **Pandas:** You can easily manipulate any type of data such as – structured or time-series data.
- **Scikit learn:** is a simple and useful Python machine-learning library.

Choosing the Right Tool

REPL:

Try out one-liners
Test small chunks of logic
Ideal for beginners or debugging

Script Files:

Build reusable applications
Automate tasks (e.g., file handling, APIs)
Use in production

Jupyter Notebooks:

Visualize data (with matplotlib, seaborn)
Combine code + explanation + results
Great for data science, ML, demos

Session-2: Test Your Knowledge- Fill in the blanks (Solutions)

- Python 2 reached its end-of-life in January 2020, and it is no longer maintained.
- Spyder is an open-source Python IDE designed specifically for scientific computing and data analysis.
- Sublime Text is a lightweight and highly customizable code editor that supports Python development with various plugins.
- IDLE (Integrated Development and Learning Environment) is Python's built-in IDE, which comes bundled with the standard Python distribution.

Session-2: Test Your Knowledge- MCQ

- What feature is NOT typically included in an IDE?
  1. Source code editor
  2. Debugger
  3. Build automation tools
  4. Version control system
- What is the main advantage of using virtual environments with Python IDEs?
  1. Improved code completion
  2. Better integration with version control systems
  3. Dependency isolation
  4. Enhanced debugging capabilities

Session-2: Test Your Knowledge- Interview Questions

1. Which version of Python is currently up to date?
2. In a Python context, the acronym IDLE stands for?
3. When you see >>> inside IDLE, it means that?
4. Is indentation required in python?
5. How to check the version of python and pip?
6. List some popular Python IDE's?
7. What are libraries and Packages in Python? Name some of them.

Session-2: Test Your Knowledge- Interview Questions (Solutions)

1. Python 3.11.3.
2. IDLE stands for Integrated Development and Learn- ing Environment. It is a simple, lightweight environ- ment provided with the default Python installation, which includes an editor and a Python shell for exe- cuting code.
3. When you see >>> inside IDLE, it indicates that you are at the Python interactive shell prompt.

Session-2: Test Your Knowledge- Interview Questions (Solutions)

4. Python uses indentation to define the structure of the code, such as the bodies of loops, functions, and conditionals.
5. python –version

Session-2: Test Your Knowledge- Interview Questions (Solutions)

6. PyCharm, Visual Studio Code (VS Code), Jupyter Notebook, Spyder, IDLE
7. Libraries: NumPy, Pandas, Matplotlib, Requests
   Packages: SciPy, Flask, Django, Scikit-Learn

Thank You!

Any Questions?